

MA378: Class Test

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1.

$$p_2(x) = \sum_{i=0}^2 f(x_i) L_i(x), \quad L_i(x) = \prod_{j=0, j \neq i}^2 \frac{x - x_j}{x_i - x_j}$$

$$f(x) = \sin\left(\frac{\pi x}{4}\right)$$

$$x_0 = 1, x_1 = 2, x_2 = 3$$

$$f(x) = \sin\left(\frac{\pi x}{4}\right)$$

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2)(x - 3)}{(1 - 2)(1 - 3)} = \frac{1}{2}(x - 2)(x - 3)$$

$$f(x_0) = f(1) = \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f(x_1) = f(2) = \sin\left(\frac{\pi}{2}\right) = 1$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 1)(x - 3)}{(2 - 1)(2 - 3)} = -(x - 1)(x - 3)$$

$$f(x_2) = f(3) = \sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x - 1)(x - 2)}{(3 - 1)(3 - 2)} = \frac{1}{2}(x - 1)(x - 2)$$

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x)$$

$$= \frac{1}{2\sqrt{2}}(x - 2)(x - 3) + (1)(-(x - 1)(x - 3)) + \frac{1}{\sqrt{2}}\left(\frac{1}{2}\right)(x - 1)(x - 2)$$

$$= \frac{1}{2\sqrt{2}}(x^2 - 5x + 6) - (x^2 - 4x + 3) + \frac{1}{2\sqrt{2}}(x^2 - 3x + 2)$$

$$= \frac{1}{2\sqrt{2}}(x^2 + x^2 - 5x - 3x + 6 + 2) - x^2 + 4x - 3$$

$$= \frac{1}{\sqrt{2}}(x^2 - 4x + 4) - x^2 + 4x - 3$$

Q.2.

$$\frac{5h^4}{384} \max_{0 \leq x \leq 2} |f^{(4)}(x)| \leq 10^{-8}?$$

$$\frac{5h^4}{384} \left(\frac{\pi^4}{256} \right) \leq 10^{-8}$$

$$\frac{5 \left(\frac{16}{N^4} \right)}{384} \left(\frac{\pi^4}{256} \right) \leq 10^{-8}$$

$$\frac{5}{24N^4} \left(\frac{\pi^4}{256} \right) \leq 10^{-8}$$

$$\frac{(5)(10^8)(\pi^4)}{(24)(256)} \leq N^4$$
$$53.06 \leq N$$

There should be 54 points to ensure that $\|f - S\|$ is $\leq 10^{-8}$

$$f(x) = \cos\left(\frac{\pi x}{4}\right)$$

$$f'(x) = -\frac{1}{4} \pi \sin\left(\frac{\pi x}{4}\right)$$

$$f''(x) = -\frac{1}{16} \pi^2 \cos\left(\frac{\pi x}{4}\right)$$

$$f'''(x) = \frac{1}{64} \pi^3 \sin\left(\frac{\pi x}{4}\right)$$

$$f^{(4)}(x) = \frac{1}{256} \pi^4 \cos\left(\frac{\pi x}{4}\right)$$

$$\max_{0 \leq x \leq 2} |f^{(4)}(x)| \text{ when } x=0 = \frac{\pi^4}{256} (1)$$